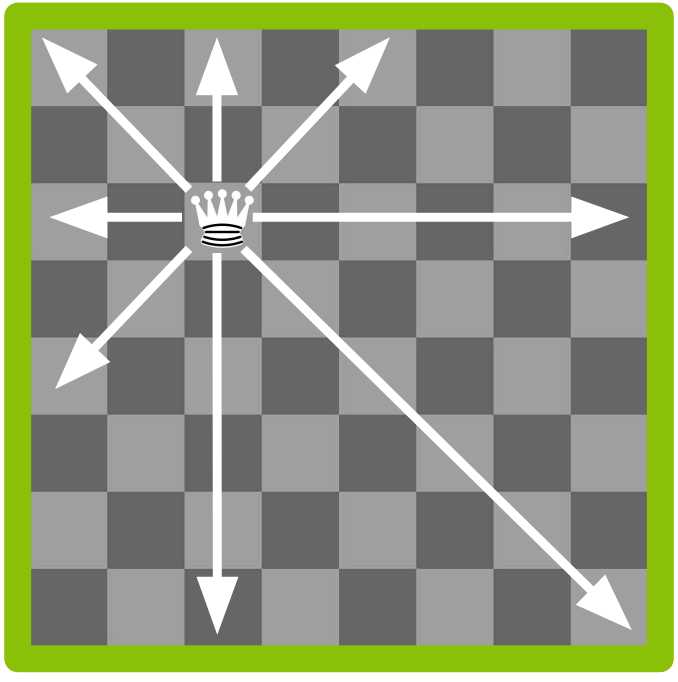


Queens of the Castle

How many queens can you fit **happily** on a chessboard?

Who is the Queen?



The queen is a very powerful chess piece!

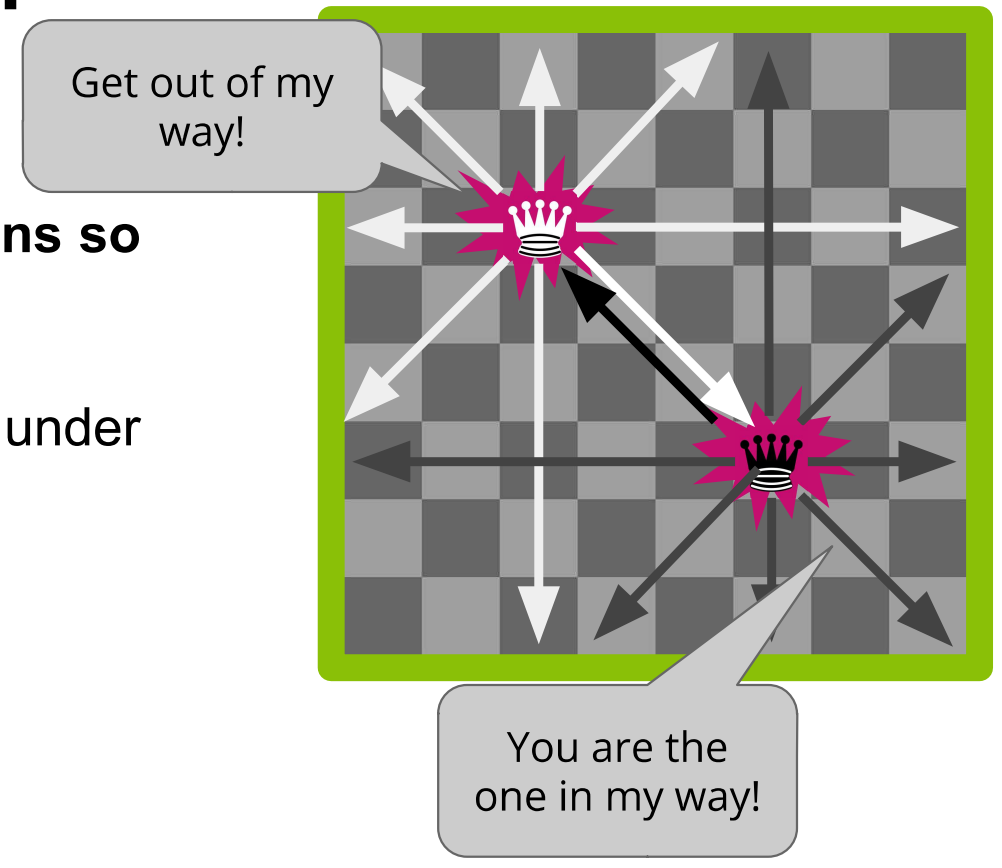
She can attack in 8 directions!
She can take on any piece in the same row or column as her. Or anything in a diagonal line to her.

More than one Queen?

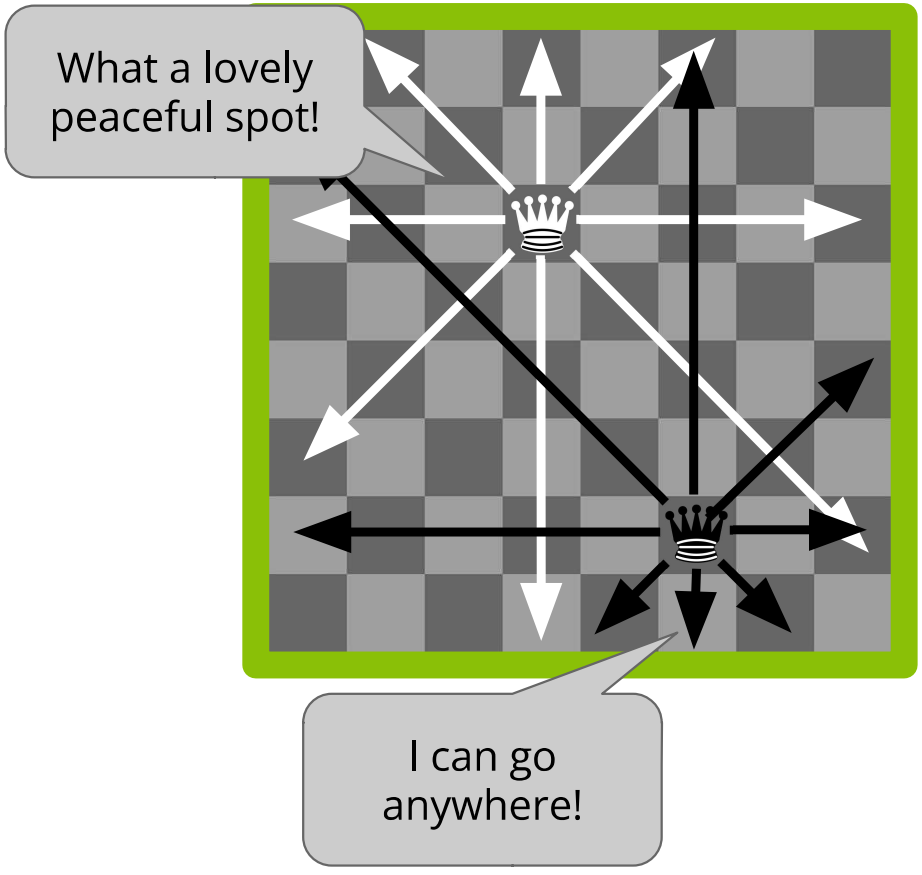
But more than one queen starts to get messy!

It's easy to accidentally place the queens so they are attacking each other.

Any piece that gets hit by an arrow is under attack!

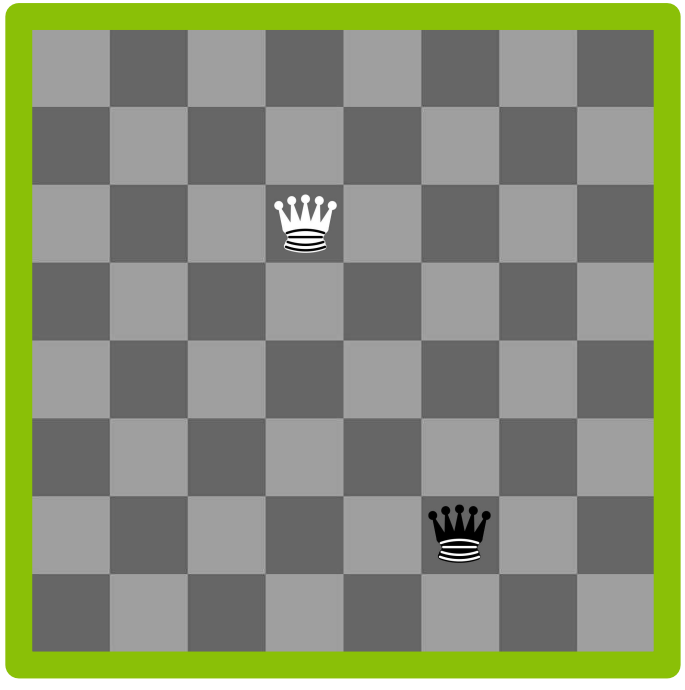


More than one Queen!



But with a little bit of effort you can make both queens happy!

Keeping Track of the Queens



I know my queens are happy now! But how do I record this solution?

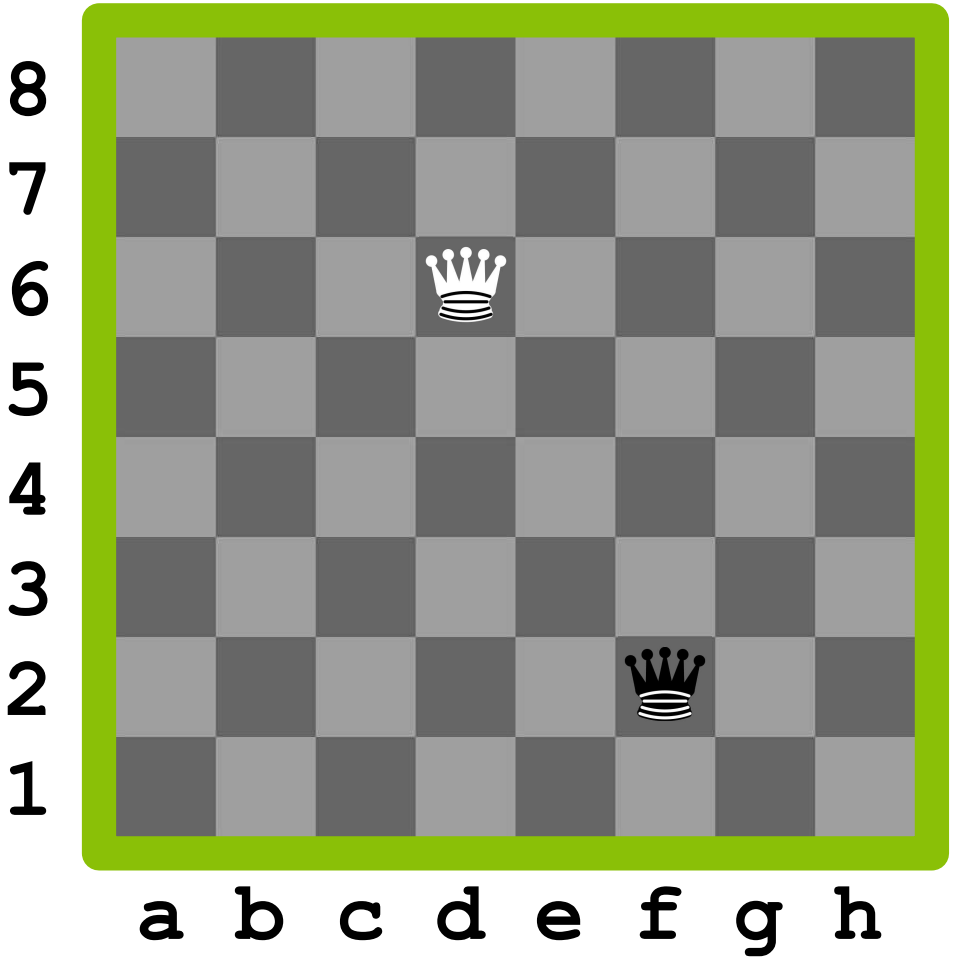
Coordinates!

Keeping Track of the Queens

Lets Label our Axis!

In chess we use numbers *and* letters!

Our x axis uses a-h, and our y axis uses 1-8



Keeping Track of the Queens

We write coordinates (*x*,*y*)

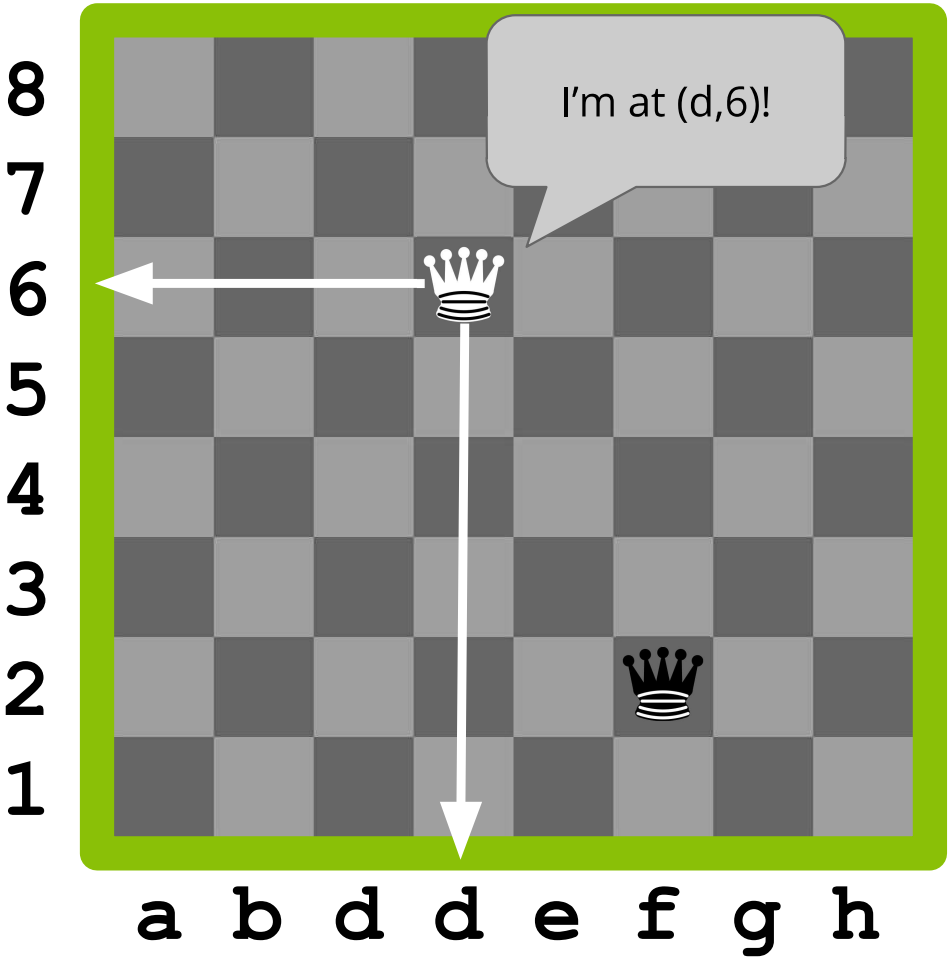
So to find the location of this queen:

1) find where it lines up on the bottom (x) axis

(d,*y*)

2) find where it lines up on the side (y) axis

(d,6)

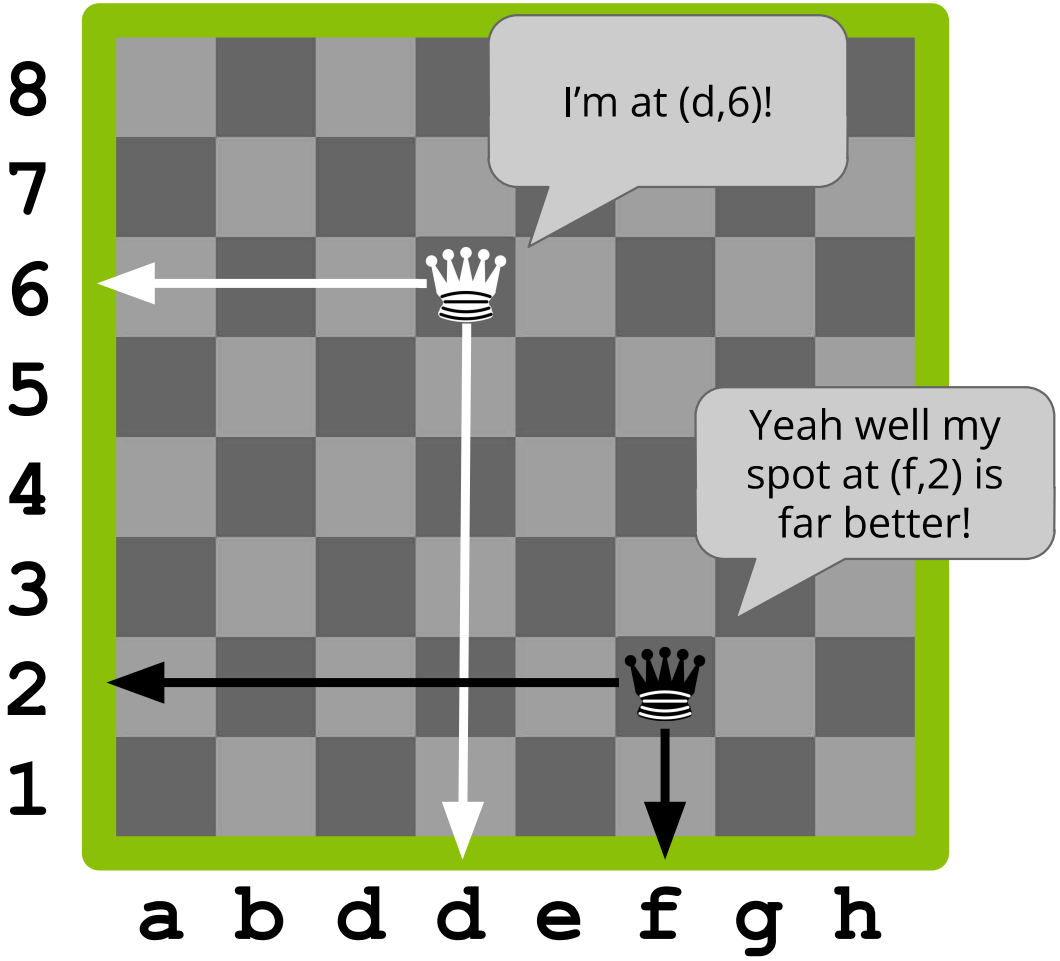


Keeping Track of the Queens

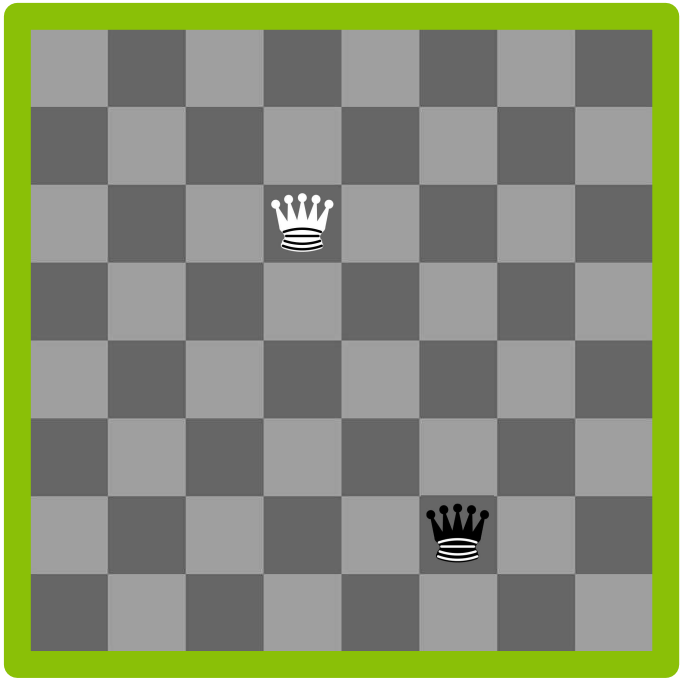
We can do this for all pieces on the board!

Then we can write out our solution for this board as:

d6, f2



Queens of the Castle



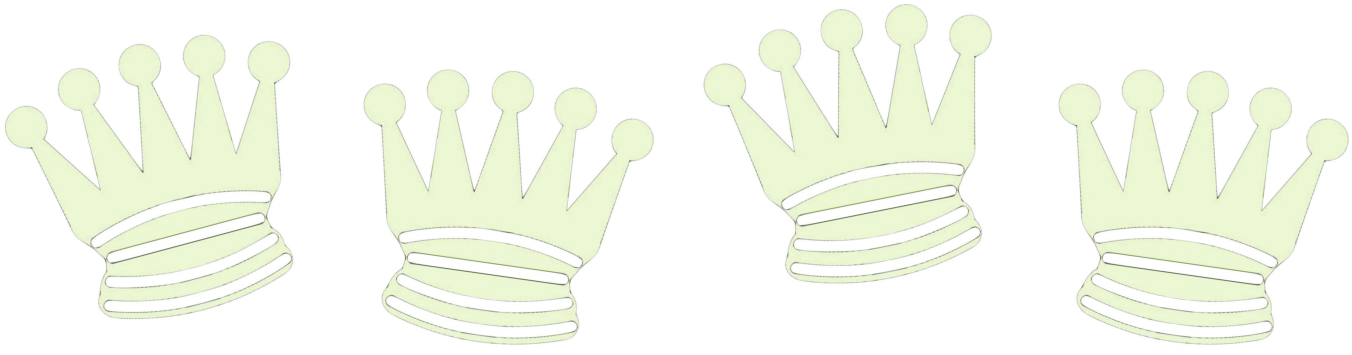
Now that we know how to arrange the board to make 2 queens happy, can we add another? How many queens can we get happily sharing?

How many queens can you get to share the board?

Can you get to 8 queens? It's very tricky!

Queens of the Castle

Alright! We aren't Queen chess pieces so we should all be more than happy to group up!

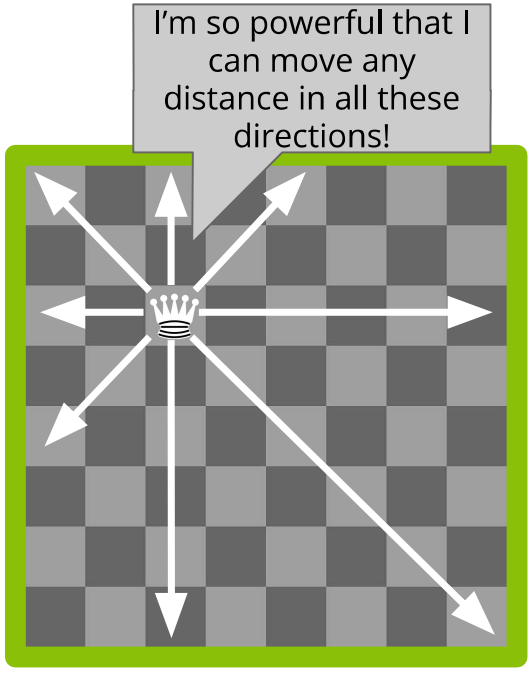


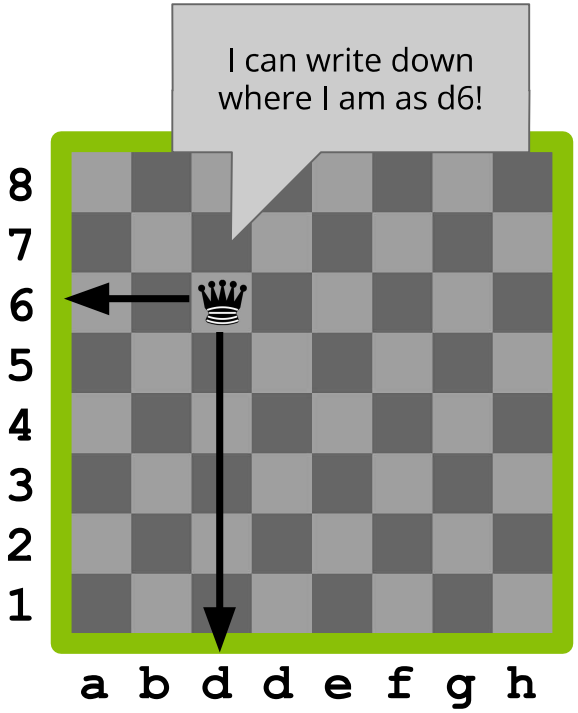
Try to get into groups of 4-5 and then send 1 member up to the front to collect your materials!

Queens of the Castle

Tip:
Use the counters to map out your possible solutions before writing them down!

The tutors have extra hints if you get stuck!





Reminder: Once you have finished the worksheet go outside to show off your answers to the tutors!